

Digital Twin Factory

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Service Traitement, Plateformes et Hybridation Aval

Physical Twin description

Coastal DT

Digital Twin Description

Earth Thematic and local Geospatial DT based on Remote sensing (Optical VHR, IRT, SAR) and high-value derived product (3D, Land Use, Change detection)

Digital Twin usages

- Predict and mitigate Climate change impacts

Engineering Digital Twin challenges

- Generation of accurate Digital Replica
- Observations and simulation data hybridation
- DT Reference Architecture

The Digital Twin Factory project

CNES with partner labs for usecases

Provides advanced and interoperable building blocks designed to facilitate the creation of digital twins.

Support research and industrial network in national and European activities.

Promote technology transfer for the development of application services based on spatial data.

Ongoing project **First demo : 12/2025**

User story

1. A partner submits a scientific challenge that could leverage a Digital Twin. They bring specific data collections and a physical model.
2. CNES provides the DTF building blocks and supports the partner in their implementation.
3. CNES guides the partner all the way to the prototype demonstration phase.
4. The partner can then independently transition to operations.

The Nokoué Lagoon (Bénin)

- Lagoon hydrological dynamics with strong seasonal effects (dry and wet season).
- Flood impact evaluation
- Pollutants transport
- High-cost HPC simulations (SYMPHONIE)
- Surrogate modeling for surface salinity and temperature

Right: Land use map derived from Pléiades & Pléiades-NEO data (CNES)
Below: surface salinity of the Lagoon over year 2020 (Okpeitcho et al., 2022)

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Metropolitan France

- Shoreline evolution
- More than 5000 km of shoreline
- Characterization and dynamics of shorelines
- Shoreline evolution modelization, adaptation strategy (what-if ?)

Top: shoreline at different times
Bottom-left: Metropolitan France view
Bottom-right: water occurrence probability

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USECASES

New Caledonia

- Major concerns linked to rising water levels (climate change)
- Detailed marine submersion simulations realised under current conditions around Noumea
- Major concerns related to measures to mitigate flood (dam construction, use of vegetation, ...)
- Other areas on the island with few in-situ data regarding land use

Top: Mesh used for the simulation
Bottom: Digital Elevation Model around Noumea

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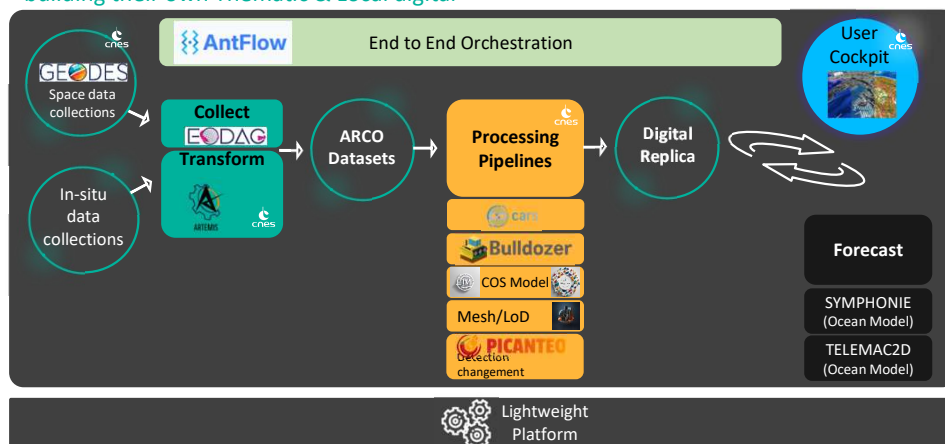
Corsica (France)

- High risks of forest fire during summer
- High dependency of fire spread speed to vegetation state
- In-situ data obtained with experimental fires
- High potential of data-driven and substitution models to partially replace empirical model used to compute the fire rate of spread

Example of experimental fire (Clements et al. 2015)
Top: post-fire aerial image
Bottom: Rate of Spread derived from aerial images

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DTF : shared toolbox with CNES advanced algorithms to support the community in building their own Thematic & Local digital



Interoperable by design

- OGC Earth Application package
 - Container
 - CWL Workflow description
- OGC API Processes
- OGC Tile and Map services
- STAC
- S3 Data access
- K8s and HPC (Slurm)
- OIDC